

# Erdős #683: reduction to prime-free intervals, and identification with the Sylvester–Erdős problem

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## Abstract

#683 asks whether  $P\binom{n}{k} \geq \min(n - k + 1, k^{1+c})$  for some  $c > 0$ , where  $P$  denotes the largest prime factor. We prove that the inequality holds automatically — for every  $c > 0$  — whenever  $(n - k, n]$  contains a prime, so the problem reduces entirely to prime-free intervals; by Baker–Harman–Pintz this leaves only  $k < n^{0.525+\varepsilon}$ . We further show that in its main range #683 is the Sylvester–Erdős problem on the largest prime factor of a product of consecutive integers. #683 itself is not proved.

## 1 Setup

By Kummer,  $v_p\binom{n}{k}$  is the number of carries in the base- $p$  addition  $k + (n - k)$ ; the set of primes dividing  $\binom{n}{k}$  is  $\mathcal{D}(n, k) = \{p : k \not\leq_p n\}$ , where  $k \leq_p n$  means every base- $p$  digit of  $k$  is at most that of  $n$ . #683 asks for  $\max \mathcal{D}$ . Throughout  $1 \leq k \leq n/2$ .

## 2 The window

**Lemma 1** (window). *Let  $p > n/2$  be prime. Then  $v_p\binom{n}{k} = 1$  if  $n - k < p \leq n$ , and  $v_p\binom{n}{k} = 0$  if  $n/2 < p \leq n - k$ .*

*Proof.* For  $p > n/2$  and  $m \leq n$  we have  $p^2 > n \geq m$ , so  $v_p(m!) = \lfloor m/p \rfloor$ . Hence  $v_p\binom{n}{k} = \lfloor n/p \rfloor - \lfloor k/p \rfloor - \lfloor (n - k)/p \rfloor$ , in which  $\lfloor n/p \rfloor = 1$ ,  $\lfloor k/p \rfloor = 0$  (as  $k \leq n/2 < p$ ), and  $\lfloor (n - k)/p \rfloor$  is 1 or 0 according as  $p \leq n - k$  or not.  $\square$

*Verified:* no exception, in either direction, for  $4 \leq n < 420$  and all  $k \leq n/2$ .

**Theorem 2** (reduction). *If  $(n - k, n]$  contains a prime then  $P\binom{n}{k} \geq n - k + 1$ , so the inequality of #683 holds for every  $c > 0$ . Consequently #683 is equivalent to its restriction to pairs  $(n, k)$  for which  $(n - k, n]$  is prime-free.*

*Proof.* A prime  $q \in (n - k, n]$  divides  $\binom{n}{k}$  by Lemma 1, so  $P\binom{n}{k} \geq q > n - k$ , and the minimum is attained by its first branch or exceeded.  $\square$

*Measured:* among the 2 248 480 pairs with  $10 \leq n < 3000$ ,  $1 \leq k \leq n/2$ , the interval  $(n - k, n]$  is prime-free for 13 909 — a proportion of 0.0062. So Theorem 2 settles 99.38% of *that* range unconditionally. The figure is range-dependent and must be quoted with its

range: the disposed fraction rises to 99.707% over  $2001 \leq n \leq 6000$  and 99.896% over  $6001 \leq n \leq 20000$ , so the residual is shrinking and no single percentage characterises the reduction.

**Corollary 3** (unconditional range). *Fix  $\varepsilon > 0$ . For  $n$  large and  $n^{0.525+\varepsilon} \leq k \leq n/2$ , the interval  $(n-k, n]$  contains a prime by Baker–Harman–Pintz, so #683 holds there for every  $c > 0$ . Only  $k < n^{0.525+\varepsilon}$  remains.*

*Measured:* the longest prime-free  $(n-k, n]$  below  $n = 2 \times 10^4$  has length 51, against  $n^{0.525} = 181$ .

### 3 Identification with Sylvester–Erdős

**Theorem 4** (refined window). *Let  $p > \sqrt{n}$  be prime with  $p > k$ . Then  $p \mid \binom{n}{k}$  if and only if  $p$  has a multiple in  $(n-k, n]$ .*

*Proof.*  $p^2 > n$  gives  $v_p(m!) = \lfloor m/p \rfloor$  for  $m \leq n$ , and  $k < p$  gives  $\lfloor k/p \rfloor = 0$ ; so  $v_p\left(\binom{n}{k}\right) = \lfloor n/p \rfloor - \lfloor (n-k)/p \rfloor$ , the number of multiples of  $p$  in  $(n-k, n]$ .  $\square$

*Verified:* 7 327 302 triples  $(n, k, p)$  with  $n < 700$  — no mismatch.

**Corollary 5.** *If  $P^+(\prod_{m=n-k+1}^n m) > \max(\sqrt{n}, k)$  then  $P\left(\binom{n}{k}\right) = P^+(\prod_{m=n-k+1}^n m)$ . Hence in its main range #683 is exactly the Sylvester–Erdős problem on the largest prime factor of a product of  $k$  consecutive integers near  $n$ .*

*Verified:* equality with no mismatch for  $20 \leq n < 500$ , all  $k \leq n/2$  with the set nonempty. *Measured:* over  $n < 1200$ ,  $3 \leq k < 60$ , restricted to pairs where the binding branch is  $k^{1+c}$ , the least admissible exponent is  $c = 0.4650$ , at  $(n, k) = (10, 3)$  with  $P\left(\binom{10}{3}\right) = P(120) = 5$ ; the extremal case is not drifting upward with  $n$ .

### 4 A ledger bound forcing $P \gg k \log k$

The reduction of §2 disposes of the range where  $(n-k, n]$  contains a prime. On the complementary range the size of  $\binom{n}{k}$  and the ledger bound each rail’s contribution, and the two together bound  $P$  from below.

**Lemma 6** (rail ceiling). *For every prime  $p$  and  $1 \leq k \leq n$ ,*

$$v_p\left(\binom{n}{k}\right) \leq \left\lfloor \frac{\log n}{\log p} \right\rfloor + 1.$$

*Proof.* By Kummer,  $v_p\left(\binom{n}{k}\right)$  is the number of carries in the base- $p$  addition  $k + (n-k) = n$ . Carries occur only out of positions  $0, 1, \dots, \lfloor \log_p n \rfloor$ , since  $n$  has  $\lfloor \log_p n \rfloor + 1$  base- $p$  digits and no carry leaves the top position.  $\square$

*Verified:* 660 046 triples  $(n, k, p)$  with  $n < 260$  — no violation.

**Theorem 7** (the smoothness inequality). *Let  $2 \leq k \leq n/2$  and put  $y = P\left(\binom{n}{k}\right)$ . Then*

$$\frac{k \log(n/k)}{\log y} \leq \Omega\left(\binom{n}{k}\right) \leq \log n \sum_{p \leq y} \frac{1}{\log p}.$$

*Proof.* For the left inequality,  $\binom{n}{k} \geq (n/k)^k$ , and any integer  $N > 1$  all of whose prime factors are at most  $y$  satisfies  $N \leq y^{\Omega(N)}$ , whence  $\Omega(N) \geq \log N / \log y$ .

For the right inequality, every prime factor of  $\binom{n}{k}$  is at most  $y$ , so by Lemma 6

$$\Omega\left(\binom{n}{k}\right) = \sum_{p \leq y} v_p\left(\binom{n}{k}\right) \leq \sum_{p \leq y} \left(\frac{\log n}{\log p} + 1\right),$$

and the term  $\pi(y)$  is absorbed since  $\log n / \log p \geq 1$  for every  $p \leq n$ .  $\square$

*Verified:* 16 510 pairs  $(n, k)$  with  $n < 260$  — both inequalities hold throughout.

**Corollary 8.** *For  $k \leq \sqrt{n}$  one has  $P\left(\binom{n}{k}\right) \gg k \log k$ .*

*Proof.* By the prime number theorem  $\sum_{p \leq y} 1/\log p \sim y/(\log y)^2$ . Substituting into Theorem 7 and using  $\log(n/k) \geq \frac{1}{2} \log n$  for  $k \leq \sqrt{n}$  gives

$$\frac{k \log n}{2 \log y} \leq (1 + o(1)) \frac{y \log n}{(\log y)^2}, \quad \text{i.e.} \quad y \geq (1 + o(1)) \frac{1}{2} k \log y.$$

Since  $y > k$  by Sylvester–Schur,  $\log y > \log k$ , and the claim follows.  $\square$

This recovers Erdős’s bound  $P\left(\binom{n}{k}\right) \gg k \log k$  from the ledger alone, with no input beyond Kummer, Legendre and the prime number theorem. The gap to #683 is the passage from  $k \log k$  to  $k^{1+c}$ : Theorem 7 is sharp in the exponent of  $y$ , so any improvement must come from a better upper bound on  $\Omega\left(\binom{n}{k}\right)$  than the rail ceiling of Lemma 6, which is attained only when every position carries.

**Remark 9** (register). **Proved:** *Lemmas and Theorems above, all elementary consequences of Legendre and Kummer; Corollary 3 additionally cites Baker–Harman–Pintz, and Corollary 8 the prime number theorem. Lemma 6, Theorem 7 and Corollary 8 recover Erdős’s  $P\left(\binom{n}{k}\right) \gg k \log k$  from the ledger alone. None is claimed to be new — Lemma 1 is surely folklore. What is recorded is the reduction to prime-free intervals and the identification in Corollary 5.* **Measured:** *the 0.62% proportion,  $c = 0.4650$ , the length-51 window.* **Not proved:** #683. **Falsifier:** *the least admissible  $c$  declining with the range would indicate no fixed  $c$  works; it did not.*